

Topic: Intersection Operation

Q. Describe the Intersection operation in Relational Algebra. What are the required conditions for applying it. Illustrate its use with a suitable example.

[DEF] Definition to Remember

The Intersection Operation ($\cap$) is a binary set operation in relational algebra that returns only those tuples (rows) that are common to both input relations. It is a derived operator (can be expressed as $A \cap B = A - (A - B)$) and requires both relations to be Union Compatible.

Intersection Operation ($\cap$)

* Binary operation -- operates on two tables.

2. Required Conditions (Union Compatibility)

Both relations must be Union Compatible:

3. Example

TOP_IN_MATH

Roll_No	Name	Grade
101	John	A
102	Emma	A
104	Mia	B

TOP_IN_SCIENCE

Roll_No	Name	Grade
102	Emma	A
105	Noah	A
101	John	B

Query: Find students with identical records (same Roll_No, Name, and Grade) in both tables.

Result: Only Emma (102, A) -- her entire tuple (Roll_No=102, Name=Emma, Grade=A) is identical in both tables.

(Note: John appears in both tables but with different grades -- A in Math, B in Science -- so his full tuple is NOT identical, and he is excluded.)

[MUST-WRITE] Must-Write Points (for 10 marks)

1. Intersection ($\cap$) returns only rows that are identical in both relations.
2. It is a derived operator: $A \cap B = A - (A - B)$.
3. Union Compatibility required: same number of columns AND matching data types.
4. The entire tuple must be identical -- not just one column -- for a row to be included.
5. If $|A| = m$ and $|B| = n$, result size $\leq \min(m, n)$.

6. Equivalent to SQL INTERSECT.

7. Used to find commonalities between two datasets (e.g., students excelling in both Math and Science).

[RECALL] Quick Recall

n (Intersection) -> Binary -> Returns Common Rows Only -> Derived from Set Difference -> Union Compatibility -> SQL INTERSECT